

Some Notes towards a Dual Model of Language Development—

The Computational System behind

a Non-INFL stage of Child Language Acquisition

Absent Minded: A Reply to Borer and Rohrbacher

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Abstract

See Dual route race models (baayen schreuder 1999). Recent studies show that it may no longer be enough to simply posit stark processing distinctions between rule-based theories of computation versus associative based computations leading to linguistic processing and formation. For instance, the demarcation line between what seems to be purely rule generated as typically aligned with (*inter alia*) inflectional morphology is very fuzzy at best, and seems to indicate that there may exist crossovers between the two modes of processing. See race models...

Background

The proposal is a simple one, but not altogether obvious. Firstly, we attempt to correlate the ‘output’ of early child utterances directly to the ‘operative level’ of the Computational System (CS)—i.e., for a state of a child’s Grammar (G), and where maturational development provides for different states ($G_1...G_2...G_3...G_4...$), $G_1...G_2...$ would be interpreted as being a direct reflex of $CS_1...CS_2$. By doing so, we feel that the CS ultimately serves as a kind of cogno-linguistic way-station upon which the child first gains access to the input. By tethering the product of G to CS, we can maintain by definition the biological null hypothesis and assume a child-to-adult continuity irrespective of whatever state the child’s grammar has reached—though, we do so not by the typical analyses which attempt to correlate the child’s underlying knowledge of G to that of the adult target grammar G_t (*pace* Borer and Rohrbacher), but rather by correlating the computational processes involved which perform the specific and modular tasks of G. Secondly, we attempt to offer a developmental response to the Dual Mechanism Model (See Pinker and Clahsen for a review). Overall, the proposal speaks to the following general question:

Under what circumstances and amongst which developmental stages of acquisition do children pay less attention and more attention to the details of their language?

What we mean by ‘attention-to’ can be linguistically defined as either:

- (i) (item-based) *statistical* ‘attention-to’: this would be more in line with cognitive, bottom-up processing that relies heavily on memory limitation, focus to attention, pattern building induction: viz., (*inter alia*) the early onset of phonemic awareness leading phonetic boundaries, word mapping toward categorical status, and possible syntactic outcomes based on the processing known as chunking, etc.
- (ii) (general) *rule-based* ‘attention-to’: this would be consistent with (*inter alia*) the paradigmatic building of the Inflectional properties of a language, syntactic formations and other formal properties of language that are non-statistical in nature.

In considering these two parts as compromising of a unifying approach to the theory, we might want to suggest that the natural development of child language acquisition is both bottom-up and top-down—bottom-up having to do in nature with that developmental state of CS which relies more on its cognitive, and not necessarily language-based, aspects of *communication*, top-down having to do with what Chomsky calls the more ‘formal’ aspects of *language* (Fitch et al.)— viz., the movement instigated by the checking of Infl-related material).

Overview

- [i] The study of syntactic development in children, for all intents and purposes, is reducible to a single-minded inquiry into how the very young child (implicitly) knows to distinguish between lexical *stems* and functional *affixes*. Hence, the overriding question burning in the minds of most developmental linguists is morpho-phonological in nature. It would seem that the child must at least know (*a priori*) the stem before she can then engage in a dual-track process by which ambient sound separation of the morpho-phonological distinction attributive to past tense is carried out, say, between the paradigmatic representation of the English word *play* vs. *play-ed* (a dual processing which provokes separation of the {*play*}-stem and the {*ed*}-affix). Similarly, we can ask what it is the child exactly knows’ when she correctly decomposes (from the very earliest MLU) only one of two seemingly similar phonological chunks taken from the ambient speech flow, say, *speaks* /spi:ks/ and *dance* /dæns/—where the former entails a decomposed [stem+affix] dual processing while the latter does not.
- [ii] Otherwise, absent this knowledge, it becomes quite convincing for the young child that (i) the pair *play-played* should represent altogether two different lexical stems, and, stored as such, reflect two distinct though relatively similar semantic notions (a single processing). In similar fashion, the two items *speaks* and *dance* could be equally misinterpreted by the child as either (i) partaking in similar morphological stem+affix processing (*speak-s*, **dan-s*), or that (ii) neither are composites (**speaks*, *dance*). Both assumptions would lead to erroneous projections of the type not typically found in the child language data—viz., whereas it is widely reported that children drop the 3/Per/Sg {s} in *speak*, children at the age appropriate syllabic [CVC] stage never drop the {s} in *dance*.
- [iii] Hagit Borer, in her 2003 *Language Acquisition* paper ‘Minding the Absence’ co-authored with B. Rohrbacher (henceforth B&R), counter-intuitively, proposes that only a **Continuity theory** can account for the ‘dual track’ feature of omission of inflection—the very avoidance of projecting the full paradigm is evidence in of itself that the child has knowledge of the paradigm: a kind of ‘knowing’ that can only be interpreted in conjunction with ‘continuity’. For B&R, discontinuity would wrongly predict (for stage-one) rather random omissions and/or commissions of the paradigm—something along the lines of what would be predicted of an ‘O(ptional) I(nfinitive)’ stage (Wexler).
- [iv] Citing from our own data, we provide case-study evidence that an exclusive non-inflectional stage-one exists during which zero-inflected declaratives and

possessives surface (e.g., *Baby have bottle* (= The baby has a bottle), *That mommy car* (= mommy's car) respectively). Given that we treat both the (nominal) possessive 's and the (verbal) third person singular +s as reflexives of the same agreement relation (between an inflected head and its specifier), the structures would be singularly represented: e.g., [IP Baby [I –agr have] bottle], [IP Mommy [I –agr Ø] car] (cf. Kayne).

- [v] Whereas our analysis agrees with B&R on the nature of the early data (*pace* Wexler's notion calling for an initial OI-stage)—in fact, in terms of data, we support much of what B&R end up claiming in their paper—where we differ with B&R is in how we go about understanding the nature of Continuity: it is our claim that the two hypotheses *Continuity vs. Discontinuity*—when understood in a manner which looks at the development of the computational system itself, and not the subsequent output—actually conflate and no longer become a viable distinction to any child language acquisition theory. What allows us to collapse Continuity in this way is our more biologically determined (and thus neutral stance) that when child language development is rightly viewed in the perspective of the 'Dual Mechanism Model' (see Pinker, Clahsen for review), one can account for the cited non-inflectional stage of early acquisition by addressing continuity of the child's internal *Computational System* (CS). (The label CS will be used interchangeably throughout this paper with the term *Language Faculty* (LF)). Therefore, rather than accepting continuity at face value as the null hypothesis, we entertain a two-prong hypothesis which directly speaks to the developing CS:

(i) a genetic-guided CS prong (Top-down)

or

(ii) a distributional learning CS prong (Bottom-up).

In terms of the typically understood dichotomy between 'top-down' and 'bottom-up' processes, we believe it is far more productive to unpack these labels in the same way we unpacked continuity and stick to questions of the growth of grammar via the CS.

- [vi] *Continuity vs. Discontinuity.* Irrespective of any putative difference in nature of the two impending accounts, and notwithstanding real and theoretically substantive distinction traditionally leveled at the two positions, we believe both views in fact share a common currency in espousing that language development is essentially a biologically determined maturational-based process by which young children (i) first tag and then separate {stem+affix} formations found in the ambient speech flow, and then (ii) proceed to build-up paradigmatic structures based on biological factors which may or may not have underpinnings to distributional learning. Regarding the biological stance, the delayed onset of neural representations by which specific language tasks (such as inflection) are preformed to the eventual development of specific cortical regions of the brain: assuming there to be a marked increase of lateralization in young children

corresponding to Broca's area (the region claimed to support 'rule-based' procedural inflection (Sahin et al.).

Introduction

- [1] The study of syntactic development in children, for all intents and purposes, is reducible to a single-minded inquiry into how the very young child (implicitly) knows to distinguish between lexical *stems* and functional *affixes*. Hence, the overriding question burning in the minds of most developmental linguists is morpho-phonological in nature. For instance, it would seem that the child must at least know (*a priori*) the stem before she can then engage in a dual-track process by which ambient separation of the morpho-phonological distinction attributive to past tense is carried out, say, between the paradigmatic representation of the English word *play* vs. *play-ed* /ple-d/ (a dual processing which provokes separation of the /play/-stem and the /dl/-affix). Otherwise, it could be conceivable for the young child that the pair *play-played* would represent altogether two different lexical stems, and, stored as such, reflect two distinct though relatively similar semantic notions (a single processing): perhaps not unlike what we do find regarding derived words where an otherwise 'two-morpheme' analysis of [*teach*]-{*er*} is processed (tagged, stored and retrieved) as a 'single-morpheme' stem [*teacher*], similar to how the word [*brother*] is stored.¹

- [2] A two-point conclusion is reached in this paper: (i) that children have instant access to and make tacit use of innate syntactic knowledge, allowing them instinctively to know to separate stem from affix—leading to a *Gradual Development Hypothesis* which shows developmental asymmetry between the acquisition of lexical vs. functional categories (Radford 1990)—and (ii) that such *prima facie* knowledge naturally arises from *The Dual Mechanism Model*, a processing model that offers the best of both worlds in that it can account for both how the child comes to 'know' (*a priori*) sole lexical stems in the first place, and subsequently, how such stems come to be distinguished and project morpho-phonological material leading to discrete stem vs. affix separation. The nature of this Cartesian knowledge we come to be expressed as an internal mechanism, seated as an intermediate-level of language processing (a layer which precedes even the role of 'stripping', prior to spell-out to PF and LF interfaces).

- [3] Finally, an interesting and potentially far reaching implication is advanced stating that there resides not only a dual routing system in the brain for the split processing of *stem+affix* material, but that the proposed dual model can be extended in such a way as to cast an entirely new dual-typology of language in the sense that (i) 'modular-complex' *weak-stem/synthetic* languages with a low, medium to high gradient range of stem modularity (English, Spanish to Hebrew, respectively) come to use the cerebral rule-based processing area as an additional language storage capacity, in contrast to (ii) 'modular-simple' *strong-stem/analytical*, agglutinative languages (Chinese, Hungarian, Turkish) which,

due to their global modularity, mitigate inflectional affixation to a much less complex system, thus preserving a more economically robust single storage capacity based entirely on the frequency-based processing area of the brain^{2,3,4}.

- [4] When this notion of modular-complexity is raised in conjunction with a previously established parameter setting dealing with [+/- Bare-Stem] languages, a new and powerful tool is fashioned allowing us to better describe and explain child functional category/feature onsets within divergent language groups. It is therefore argued that many of the cross-linguistic asymmetries found in Early Child Inflectional Development should be better thought of as reflecting how the innate Language Faculty provides languages the selection of a single vs. a dual storage capacity.

Assumptions and Suppositions

- [5] We follow Chomsky's (1995) 'The Minimalist Program' (MP) throughout and assume language has real structural antecedents to the mind/brain—*The I-language is a state of the mind/brain*. We then assume there to be real physiological connections between language, syntactic structure and relevant brain processing. A 'theory-of-mind' must therefore interest us as we build upon any language theory. Still, very little is known about how a brain bootstraps itself and creates a mind, as there is little understanding of the brain/mind relation. An isolative 'theory-of- brain' and its processing, however, is much better understood and allows us to be a bit more precise. There is now strong scientific evidence to suggest that innate architectural principles of the brain process linguistic information in two fundamentally different ways, thus, by extension, determining how lexical vs. functional features project into syntactic structures. We likewise assume that this dual processing, as defined and expressed (overtly/covertly) in 'syntactic trees' has real linguistic relevancy and is constrained by certain legibility conditions (at PF/LF interfaces). We further assume this relevancy captures a natural cut, or a language divergence, as characterized by a dichotomy between *semantic-based language* (pertaining to the VP) vs. *syntactic-based language* (pertaining to the IP).
- [6] Following the general framework as laid out in Chomsky 1995, along with crucial assumptions as detailed in 'Distributed Morphology' (DM) (Halle and Marantz 1993) (viz., the notion that functional heads serve as the locus of lexical insertion, with word formation occurring in the syntax as a result of the syntactic combination of such heads, and, that due to maturation, functional heads may go unspecified in the syntax) we assume this cut shows itself in the lexicon as follows and may have a real brain-language physiological correspondence to how a *higher syntactic* position (IP-dominated) versus a *lower syntactic* position (VP-dominated) respectively captures the *overt vs. covert* checking of features:
- [7] The V(erb) P(hrase) contains phono-semantic features. These features are substantive in some way, and thus interpretable at each relevant interface. Lexical knowledge associated with the VP is therefore said to be unavoidably trivial and instinctive to a certain degree, as derived by a set of *data-driven* 'sound-meaning' relations appropriated to the PF/LF interface. Features of the VP are labeled

- +Interp(retable) and, as a rule, do not require the movement of its head into a higher functional projection. DM labels such features as l-morphemes (lexical). They are idiomatic and correspond to concrete categories of Verb/Noun/Adjective. We will label +Interp features as sensori-motor/semantic.
- [8] The I(nfleclional) P(hrase) contains features that are not substantive in some way and are therefore -Interp at either phonological or semantic interface. Functional knowledge associated with the IP is therefore said to be non-trivial, as derived by a set of *rule-based* relations which must then be ‘checked-off’ at the Phonological Form (PF) & Logical Form (LF) interface. Whenever the typology of a language forces -Interp features to be checked in the overt syntax, this forces the features to project in the morphology of the language. We will label –Interp features as recursive/syntactic.
- [9] Whenever such features can go unchecked in the overt syntax, the choice between overt vs. covert expression in the language depends on that language’s specific morphological parameterizations. (See §35 Typology). These formal features represent for the most part what drives language variation. The reasoning for their existence in language remains somewhat of a curiosity (it may be that such non-substantive features ultimately drive the unique ‘displacement property’ of human language—in the very general sense that functional checking motivates movement up the syntactic tree). DM labels such features as f-morphemes (functional): they are non-idiomatic and rely on vocabulary selection.
- [10] Following more precise thinking about this dual architecture, as it relates to what is often called the *Language faculty* (Lfac) (see Hauser, Chomsky and Fitch 2002 for discussion), we find that while, indeed, sensory-motor systems are involved with crucial aspects of language that control much of the *Saussure* ‘form-meaning’ exchange of language, this system or component of language is maintained by a cognitive underpinning and thus is, in of itself, insufficient to carry out higher order, formal operations such as recursions, a potentially (non-cognitive) non-local displacement operation (i.e., syntax). (We can take some stock in this by our affirmation of a language vs. cognitive double disassociation, as expressed by proponents calling for an autonomous Lfac. (See Smith and Tsimpli for some discussion of this double disassociation.) What we precisely mean when we say there is a dual, modular Lfac amounts to little more than what is usually meant behind general notions prescribed to the DMM itself: viz. that the Lfac converges on two very discrete areas of processing to the extent that such processing leads to accounts of language acquisition and language use.
- [11] This first area of Lfac could be roughly described in traditional cognitive terms—alluding to the aforementioned rift between cognitive vs. linguistic processes—as having a more-or-less bottom-up treatment of language insofar as memory, analogy, common-sense, abduction and associative pattern finding, etc. are utilized in the account—‘a cognitive/sensori-motor component’ of language. This second area, which establishes rule-based syntax, is then offered only as an additional overlapping layer onto the first processing. (It is this second processing which is labeled as the ‘recursive component’ of Lfac by Hauser *op. cit.*). (Nb. The fact that I use the term *overlapping* here should in no way be interpreted as

meaning non-essential. Quite the contrary: it is my position that without this overlapping recursive component (albeit a component that finds itself straddled to lower-level cognitive underpinnings), by all definitions, human language (L_{fac}) would not exist).

- [12] In an attempt to pin this ‘dual L_{fac}’ (henceforth described as the ‘Full L_{fac}’ keeping to the spirit of Hauser et al.’s terminology) to that of aforementioned discrete categorical processes, we might wish to suppose thinking along the lines that follow (permitting me here to simply interchange the terms ‘L_{fac}’ with ‘brain’):
- [13] Suppose that the VP is exclusively pinned to the *sensori-motor component* of the brain, the more primitive part of the brain that controls cognitive motor-skill, memory, lower-level associative learning (*conceptual-intentionality*.) In this sense, the VP presides as the lexical category par excellence alongside semantic-based [+Interpretable] features. In this sense, lexical categories exist prior to any computational numeration. We’ll call this part of language processing ‘sensori-motor language’ (SM_L).
- [14] Suppose, in addition, that the IP is rather pinned to areas of the brain which house more abstract levels of thought (perhaps exclusively relating to the left/frontal lobe region of the brain). Functional categories are the result of the combination of a lexical item and a functional/inflectional feature—a *recursive component* leading to a numeration of sorts (*computation*). Hence, we define the full LF as that which converges upon two language components: (i) the sensori-motor component, and (ii) the recursive component. We’ll call this type of language processing ‘recursive language’ (R_L).
- [15] Suppose further that classic processing distinctions between *Derivational* vs. *Inflectional* morphologies may be fuller understood in light of this dichotomy, i.e., that derivations ultimately trace their origins back to interpretable features internal to the lexicon (l-morpheme), and, conversely, that Inflections are nothing more than ‘morpho-phonetic fillers’ (expletive in nature), the remnants of some computational operation of a broader syntactic scope (f-morpheme). It is this latter observation *inflection* that will interest us in this paper. More than any other single linguistic property, it is the nature and projection of inflection that gives a language its typology—languages differ in inflectional systems.
- [16] We will have more to say below in the following section regarding how I believe this very limited and prosaic interface leading to SM_L may actual aid the child in eventually developing an accurate target grammar—keeping to the spirit of the twin arguments ‘the importance of starting small’ and ‘the less-is-more hypothesis’ (Elman, Newport).

An Intermediate-level way-station holding input

- [17] One leading tenant of the present proposal is the idea that there exists an ‘intermediate-level’ processing mechanism (much like a way-station or storage capacity and positioned prior to the stripping of spell-out for PF, LF), whose

- sole task is to first search, screen-out, and potentially build-up lexical/functional structure leading to a good approximate of the adult target grammar. The final stages of this processing could be consistent with what is commonly understood as *UG Principles & Parameters* interacting with the data; but the point here is that there is some prior level which holds the language input in place so that UG processing can take place.
- [17] We suggest that the brain (Full Lfac) may house two separate language storage capacities, perhaps in some way paralleling PF vs. LF representational systems (Phonological vs. Logical Forms) (prior to stripping): the former involving, thought perhaps not exclusively, the sensori-motor component (morpho-phonology), and the latter involving the recursive component (syntax). In the main, we believe the recursive component would also involve notions relative to UG such as parameter-settings, reflecting to some degree how a language selects to show these fillers—namely, (i) a language may opt for fillers as ‘true’ affixes, disjoint from the lexical stem, or (ii) opt to incorporate the fillers somehow as part of the stem. Following this assumption through, it then can be argued that it is ultimately the recursive/syntactic component of language that first appropriates and selects for a morphology type—syntax drives morphology. (In generic ways, the former entails the operation *Move* (a PF component), while the latter entails the operation *Merge* (an LF component). In more concrete ways, the former operation *Move* may force an entire lexical item to overtly raise with an affix, as seen in so called *pied-piping*, (saving the operation at the level of PF) since a formal affix unattached to its stem would crash at PF). In these terms, *move* (at PF) is morphologically driven, while *merge* (at LF) is an operation which drives syntax.
- [18] In another crucial way, these two separate storage capacities may likewise play a role in arguments surrounding the actual computation of linguistic material, and how this computation should be presented and understood via structure and notation. I am particularly thinking about specific distinctions typically understood in terms of how one defines ‘syntactic nodes’ from ‘feature projections’, whereby the former is defined by lexical interpretable material and the latter is defined by non-interpretable material. (Much of what has come out of the Minimalist program has been to question the status of syntactic nodes over feature projections—e.g., it is instructive to simply take a cursory look at the final pages of Chapter 4 of the Minimalist Program to see just what types of trees we find ourselves drawing, anyway. It doesn’t take long to see that much of the phrase structure has been reduced to +Interp substantive nodes—as evidenced by the preponderance of VPs, *light vPs*, D, T, and C—all of which purport to having various forms of semantic +Interp material. It is noteworthy to remind ourselves that it was in this sense that Agr got deleted altogether as consisting of a node: (the impoverished Agr being relegated to an –Interp feature projection in order to save theory internal derivations). It is further noteworthy that in pre-Minimalist terms, or, at least in the eyes of more traditional grammarians, it was considered that syntactic nodes should only be exclusive projections of lexical categories, and nothing more (e.g., NPs, VPs, AdjPs, PPs). There are some parallels to draw on here, I think, now with the lexical category being supplanted by +Interpretability.

- [19] Hence, as found in current analyses, we seem to be widening further this cut between interpretable and non-interpretable features—as it is captured by the cut between Tense (now, a seemingly ‘sensori-motor component’ dealing with semantics which forms its own node) and Agreement (a seemingly recursive component which is ‘reduced’ to merely residing as a feature within a larger given node). (See Chomsky 1999). To the extent that T holds semantic +Interp features, Chomsky then concludes that T must therefore be a substantive lexical category rather than a functional category, as had been assumed up until then).
- [20] Expanding on this argument, a dual-typology of *Semantic transparency* vs. *Grammaticization*-based languages is suggested, termed herein as a ‘Modular-simple’ vs. ‘Modular-complex’ parameter (See § Inflections and Typologies). The parameter reduces to the selection of how a given language stores and projects such vacuous filler material. Nothing hinges on this last supposition: the dichotomy could be upheld irrespective of whether or not such a dual storage capacity is ultimately correct. Future developmental research will surely look toward MEG, fMRI and PET scans to ultimately replace models of syntactic tree diagramming. Only through better brain-to-language modeling will we eventually transform language theories into biological certainties.

The Sensori-motor & Computation Component Interface

- [21] There is a fair amount of good reason today to suspect that such an *interface* between these two sides of the Lfac actually may serve as a way-station, a kind of **input holding platform** for the child in the course of her language acquisition. In this sense, it would be claimed that the child draws from this inter-level database of input in generating her first approximation of the language, either having to do with abduction references, analogy of paradigms, and/or parameter settings. Specifically speaking, one possibility here might be to say that the classic argument over the *poverty of input* is not specifically due to the poverty of the (full) target input *per se*, but rather that the poverty is a direct result of how the interface interacts with and holds the input: this could be explained by assuming that one operation of the interface is (i) in the first stages to seek out only that ‘limited’ language which is sensori-motor identifiable and (ii) partition it to the child’s Lfac, thus presenting to the child the limited input as if it were the full input. This, of course, would lead to an impoverished early stage of acquisition, but not for reasons which are typically attributed to the impoverished target input *per se*, but rather attributed to this interface holding platform which, in first course, only makes available to the child SM_L.
- [22] At first approximation, the notion of ‘a poverty of stimulus’ being thus the result of maturational, subjective innate dispositions (rather than or in addition to the actual objective stimulus) has been a position I have found long held in the literature:

As Children, we learn to decode the chaotic messages
which meet us from our environment (*an objective*

poverty). We learn to sift them, to ignore the majority of them, and to single out those which are of biological importance for us either at once, or in a future for which we are being prepared by a process of maturation (*a subjective poverty*). (Karl Popper p.63 *Objective Knowledge*. Emphasis belong to me.)

- [23] The upshot of this may be to say that there is not only one type of poverty of stimulus—the sort Chomsky addresses—but that there are two: (i) the objective/external poverty (attributed to Chomsky 1965) and (ii) the subjective/internal poverty (as suggest by Popper).

The ‘Less-is-More Hypothesis’ as Result of the Interface

- [24] Expanding on this idea of an internal and subjective poverty of stimulus—as it has to do with child language acquisition—let us turn our attention to some more recent findings which seem to support the position. Elman and Newport’s ‘Less-is-more Hypothesis’ provide a mechanism by which an account for the ‘starting small’ can be made. In following their argument, we likewise assume that children start out without rules and paradigms. (See note 13.). Only later are they said to *construct* their paradigms incrementally, one feature at a time, given the onset of LFT activity. (Nb. Janet Fodor has taken a similar tac in suggesting that *parameter setting* is not activated by the full/sentence structure of a given language—given that the child could not possibly handle the total load of parameter selection the full structure would entail—but rather children gain access to parameterization first via smaller partial treelets of structure). It may be that such inflectional languages which start out ‘small’ may otherwise use mechanisms available in other non-INFL languages (though still provided by UG) to mark notions of T/AGR. (See Hyams (1996) for a semantic treatments of otherwise null INFL markers.).

Child Language and Discontinuity

- [25] It is now well documented that English children pass through a (gradual) developmental stage in which they comprehensively fail to project functional categories and features. At the same time, it is equally well documented that children of other language groups, relative in age to their English counterparts, may not proceed in a similar (gradual) fashion in omitting the same formal categories and/or features. Thus, it would seem any attempt to tether formal functional deletion to a more general cognitive deficit, as based on a biologically-driven maturational theory of language development, would run into several difficulties.
- [26] It is argued in this paper that a unifying heuristic procedure which incorporates a two-prong analysis of (i) *Feature Complexity* and (ii) *Inflectional Paradigm Complexity* indeed maintains, as the working null hypothesis, a biologically based

maturational account and may help determine the eventualities of a language's development, as attested in child language acquisition. The heuristics come to detail a 'Converging Theories Hypothesis' of sorts which attempts to describe language acquisition along the lines as being both 'Discontinuous' in nature to that of the adult target language (regarding a top-down paradigmatic/structure analogy, thus saving some elements of a maturational account), and at the same time 'Continuous' (regarding type-token frequency learning which is described as a bottom-up cognitive universal). What we mean by simultaneous 'discontinuous' & 'continuous' can only be understood via the defined 'Full (two-state) Language Faculty' (FLF) as presented above: (i) a *sensori-motor faculty* (SMF) and (ii) a *recursive-computational faculty* (RCF). (Similar understanding of the FLF also has some bearing on discussions concerning language as modular (e.g., Fodor vs. Pinker)).

- [27] The 'Dual Mechanism Model' (DMM) is then advanced as being best positioned to account for the attested child language asymmetries found amongst the diverging language groups discussed. Therefore, it is to be maintained that only insofar as the FLF is accepted as the total language operating system can we then postulate for a maturational theory of language development.

The Literature

- [28] It is now well documented that English children pass through a (gradual) developmental stage in which they completely omit functional [categories] and {features} such as [CP, IP, DP] and {T(ense), AGR(eement) CA(se)}. Following the developments of research models over two decades, as advanced in Felix (1984), Borer & Wexler (1987), Guilfoyle & Noonan (1988), Radford (1990), Wexler (1994), Hyams (1996), and Radford & Galasso (1998), there is today strong, surmounting evidence to support some kind of *Discontinuity* (or, for a lack of a better term, some 'non-adult-like' conceptual linguistic formation) between the English speaking child and adult, at least with regards to the aforementioned projections of functional categories and/or their features. (See §21 for some data).
- [29] After the attested 'No Functional Categories' stage, children then seem to pass through a slightly more developed (albeit non-target) stage in which they confuse much of the functional grammar they have acquired. For example, it is not uncommon for English speaking children to go through an early stage of development in which they over-regularize a Gender feature, whereby the pronoun He is used for both males and females alike (CHILDES, Brown files: 13). In addition to a gender pronoun feature non-specificity, a child might confuse the 1/2nd person feature spell-outs of the pronoun I for You and You for I, or may simply generalize the 3rd person possessor features for possessive {'s} (Tom's) onto the first and second person possessor (*my's*, *your's*) (Chiat 1982).
- [30] In addition to some of the feature confusion or non-specificity as mentioned above, the most salient fact about child language is the systematic 'omission' of functional material—a 'No Functional Category' stage—both in the forms of the functional categories themselves, as well as with the associative parameters that maintain such feature specificity (Spec-Head relations in higher-order functional

phrases).

- [31] Notwithstanding such apparent data which seemingly call for *Discontinuity* between child and adult grammar, a split has emerged nevertheless amongst those developmental linguists (i) who espouse for an initial stage of child syntax which begins in *complete absence* of functional material (a Non-Functional/Inflectional stage as suggested in Radford 1990/Radford & Galasso 1998) *contra* those (ii) who espouse for an initial stage which begins with *partial absence* of functional material (an O(ptional) I(nfinitive) stage as suggested by Wexler 1994). Proponents calling for such *Continuity* tend to cite not only English as one potential confirmation source (leading to the aforementioned rift between Radford vs. Wexler), but cite the early emergence of functional categories in other language groups, thus complicating Radford's general claim for an initial No-Functional stage-1. New criteria can be established in order to make the debates more precise. Questions into whether or not a language's inflectional system is sufficiently rich to spawn very early MLU onsets of affix morphology enter into the equation.
- [32] This then naturally leads to question regarding whether or not the early presence of syntactic movement is linked to the checking of functional material (Chomsky 1995). One might ask whether the given language manifests a kind of deviant inflectional usage (as attested by abnormal Aphasia/Specific Language Impairment studies)—many such errors in certain language types essentially amount to errors with wrong inflectional usage of 'commission' as opposed to any complete inflectional 'omission' (in defiance of the discontinuity theory as mentioned above). (We'll consider these each in turn in §11 below). Hence, the two differing hypotheses, Discontinuity vs. Continuity, could be said to converge roughly at the cross-roads of morphological typology: namely, it has been suggested that when a language has a relatively rich inflection [+INFL], that language will seemingly exhibit the early projection of functional categories/features (as attested by child language studies of the relevant language). However, when a language does not have a sufficiently rich inflection [-INFL], that language will exhibit a somewhat delayed projection of functional categories/features (a delay that has been cited as providing general evidence for maturational based hypotheses). It is in this general sense that language morphological typology is said to drive (top-down) the nature of early child syntactic projection (in contrast to, say, a biologically determined maturational process). Put another way, (i) it is first *a priori* knowledge of morpho-phonology that triggers the appropriation of syntax, (ii) the syntax, in turn, then seeks out both the *choice of words* and the *nature of the paradigm*.⁵
- [33] Hence, following the DMM, children 'know' that Inflected forms *are not* lexical since such forms are conceptualized and generated in a separate processing modular. Following our discussion of the FLF, The DMM maintains that a clean separation takes place between the *lexical stem* and the *affix*—a stem is 'meaning based' and thus housed in the temporal-lobe region of the brain (Wernicke's area), while the affix is 'abstract' and thus housed in the frontal-lobe region (Broca's area). The lexicon lists only lexical items (stems) while functional items

(affixes) are added at a second stage in the numeration. (See §44 for diagram). The DMM credits the Brain/Mind with having two fundamentally different cognitive modes of language processing—this dual mechanism has recently been reported as reflecting inherent qualitative distinctions found between (i) regular verb inflectional morphology (where rule-based *stem+affixes* form a large contingency) and (ii) irregular verb constructions (where full lexical forms seem to be stored as associative chunks). The *Language Faculty* thus provides us with two ways of symbolic representation.

- [34] The overall approach here is therefore ‘top down’ since there is no sense in the word ‘Knowing’ unless there is first ‘Access’. (Though it is generally accepted in *Feature Theory* (Chomsky 1995) that syntax is driven ‘bottom-up’ by features which percolate up from the lexeme and enter into a phrase configuration—presumably a spec-head configuration under a movement analogy termed *Merge*—the proposed model here differs only to the extent that we believe the initial ‘Access’ to such features is ‘top-down’). This has the flavor of saying that the debate over continuity may be misplaced and badly spelled-out, and that the crucial debate rather hinges on how the maturing, neurodevelopment of the brain processes (or doesn’t process) the ‘accessed’ paradigmatic/inflectional material generated by the two diverging language groups. Having said this, I follow Meisel (1994) and suggest that it is the full-fledge realization of the INFL-paradigm that should be the real measure of competency, rather than any language specific piece-meal affixation process by which morpho-particles affix to stems, since the latter could always be explained away as misanalysis on the child’s part in ways that do not directly speak to a well formulized inflectional paradigm (e.g., formulaic and/or lexical incorporation, lexical redundancy rules, un-analyzable chunks, etc.).⁶
- [35] *Early Inflection: Pro-Continuity.* Contrary to what some might like in assuming child-to-adult discontinuity—viz., that all languages start out with deficient functional categories, say, owing to a general cognitive development of functional categories—a good amount of data found in the literature calling for a least some aspect of early functional competence include language groups such as French (Pierce: 1989, 1992) and German (Clahsen: 1990, 1994).⁷ Hence, those clinging to continuity can always claim that at least one functional category is present from the very earliest MLU of child speech (hence, quieting any clarion call for a strong discontinuity hypothesis). For instance, Pierce shows that from the early age of 1;9-2;3 (Nathalie files) +Fin(ite) V(erbs) (hosting T/AGR) show movement via their raising out of VP above NegP and into a functional IP, whereas -Fin(ite) Verbs remain VP in-situ. (NegP is traditionally viewed as being an intermediate phrase situated between the lower VP and the higher IP). Such findings illustrate syntactic sensitivity on the part of the child by the fact that the features in the lexeme drive syntactic movement for purposes of checking. (The checking of formal -Interpretable features such as CA and AGR must ensue under concord between a functional Spec-Head configuration).

- [36] *French:* IP (Pierce 1992) +/-Finite Syntax (Pierce 1989)
- (i)
-
- [+Finite] [-Finite]
- | | | | |
|------------------------|-----|----|--|
| <u><i>pas</i> verb</u> | 11 | 77 | |
| <u>verb <i>pas</i></u> | 185 | 2 | |
- (a) veux_i pas t_i lolo => Functional projection
 [1Sg Pres] want not water
- (b) Pas manger => Lexical projection
 Not eat [Inf]

- [37] *German:* FP (=Functional Phrase) (Clahsen et al. 1994: p. 9)
- (ii)
-
- [+F] [+V]
- | | |
|--|---|
| <u>[+Fin] V2 utterances</u> | <u>[-Fin] V-final utterances</u> |
| (a) Kann er nich (2;9)
(can he not) | (c) hase auch auytofahrn (2;4)
(hare also car drive) |
| (b) ich hab hier reintecken tasche (2;6)
(I have here put in bag) | (d) mone auch lump ausziehen (1;11)
(Simone also rag take off) |

Clahsen sites very early MLU German child sentences in which 90% of all Finite Verbs show raising across negation into a higher host, verb-second position (here, labeled FP (=Functional Phrase)). Non-finite verbs in V2 position are practically non-existent (op. cit: 13).

- [38] *SLI: Pro-Continuity*. In addition to the question of the early (vs. late) projection of inflectional, a second line of inquiry has begun to look at Specific Language Impairment (SLI) as a means of teasing out just how the morpho-syntax within divergent language groups is processed. For example, one factor that arises out of morphological typology is the classification of a matrix parameter that holds between [+/-INFL(ectio(n)al)] to [+/-Bare Stem]. For instance, languages which have a rich inflection [+INFL] tend to also have a [-Bare Verb Stem] parameter setting (Hyams 1987) and *vice versa*. Languages such as Italian, Spanish, among others (with perhaps the most severe case being Hebrew) are non-stem-based languages in the sense that the projection of a bare stem doesn't constitute a word. (In the case of Hebrew, the radical root of a word can only be realized and projected in conjunction with the root's morpho-phonological paradigm). Whereas it is grammatical to project the bare verb stem *speak* in English (as in the infinitive usage *I can 'speak' French*), it is ungrammatical to do so in Italian or Spanish **parl-*, **habl-*. What turns out to be interesting along these line of inquiry is that with respect to both child language development and SLI, [+INFL]=>[-Bare Stem] languages tend to show immediate and systematic functional category/feature projection on the one hand and only SLI inflectional error usage on the other. (See Grodzinsky (1990) for a detailed look into Aphasic/SLI studies). For example, Italian SLI subjects never omit inflections as that would violate word-structure properties. Italian SLI rather typical involves inflectional errors having to do with misplaced Gender, Number and Agreement—e.g.,

- (i) *Quest-o macchin-a
 This {masc} car {fem}
 (This car)

- This contrasts with both normally developing children as well as SLI children of the matrix correlate [-INFL]=>[+Bare Stem] (i.e., English) who show a stage in which functional categories/features are entirely missing. What this seems to suggest is that before such language specific data can outright challenge the general notion of discontinuity in child language acquisition, a more detailed analysis of the language's morpho-syntactic processing must be obtained.
- [39] Since French does constitute as a [-INFL]=>[+Bare Stem] language, how do we square this with the above observation made by Pierce that there is very early functional projection? One would expect that since French constitutes a weakly inflected though synthetically strong language, discontinuity theorists would assume an initial stage in which inflection is not yet procured. (For further arguments along this line of assumption, see §35 **Inflections and Typology**).
- [40] Well, as it turns out, there is some evidence to suggest that the picture is somewhat mixed and that Pierce's French children (perhaps more so than Clahsen's German Children) do have a stage (albeit short lived) during which they completely omit functional categories—'a clarion call for discontinuity'. For instance, as Radford notes (1995) and as pointed out by Atkinson (1995 p. 53)

...‘if we read [Pierce’s] work carefully, we uncover the observation that in the very first recording of Nathalie at age 1;9;3,..., Nathalie uses only nonfinite verb forms, not finite verbs.’

Atkinson notes: *Radford goes on to cite Pierce’s own observation of a very early stage at which inflected forms are absent. Though it is unclear what Pierce precisely means by ‘uninflected forms’, we can at least assume that she has identified a stage in early French which shows no systematic finite/nonfinite distinction.*

- [41] Similar conclusions could be reached for German and Dutch (Wijnen, Bol 1993, p. 247, quoted in Atkinson). (*Italics* belong to Wijnen & Bol):

‘...there are indications that during the first phase of grammatical development, Dutch (and German) children only use nonfinite verbs...’. This implies that the ‘Optional Infinitive’ stage may in fact not be the very first stage of syntactic speech and that a prior stage exists which shows No Inflection—a Non-Inflectional stage.

- [42] When we turn to Spanish, however, we see a somewhat more complex picture emerge. In light of the fact that Spanish is a [-Bare stem] language (and thus must assign some affix material to its stem), other syntactic considerations must be applied in determining whether or not there is true functional awareness. For instance, Grinstead (2000) argues that there exists a stage-one in early child Spanish that exclusively manifests null subjects and that a correlation holds between this exclusive ‘null-subject stage’ and the specific absence of particular affix agreement inflection having to do with Tense and Number. In brief, Grinstead concludes that it is the simultaneous onset of tense and number in the inflection paradigm that ultimately triggers a particular aspect of UG pertaining to the CASE of overt subjects. The particular onsets of such inflection are complicated and potentially undermined by the possibility that simply because a young Spanish child shows affix material (infinitive endings *-ir*, *-ar*, *-er*), the child may simply be realizing these as parts of the stem due to the [-Bare stem] parameter (i.e., they are rote-learned chunks). If indeed this is the case, other factors must be considered in concluding that functional categories have emerged. A similar conclusion could be reached regarding early Italian verb+affix constructions. For instance, the underlying structure of initial inflections (e.g., *mett-o*, *mett-i*, *mett-e* ‘put-1sg/2sg/3sg’ (Guasti 1993/1994) may, in fact, be (semi)-rote-learned since there is no syntactic possibility of a bare stem production.
- [43] In addition to the bare-stem factor, it may be that when such inflectional sequences are strong, with high frequency attached to their production, the clusters may take on stem-like qualities. (In this sense, weak clusters provide better material for paradigmatic formation). The fact that subject-verb agreements are consistent may rather speak to broader morpho-phonetic notions having to do

with a kind of grammaticalized lexical incorporation to the extent that verbs ending with *-o* derive first person, etc.) It is equally important to note that Wexler's OI-stage doesn't manifest in pro-drop languages such as Italian or Spanish. One might ask why this is. It appears that the lack of OIs in pro-drop languages can be traced to properties of the AGR and INFL paradigm systems in the language. For instance, AGR in such pro-drop languages doesn't have the array of -Interp(etable) features that otherwise drive DP-subject projection and movement (via the overt checking of the -Interp subject D-feature, presumably CAsE). In this sense, it is understood that AGR in pro-drop languages rather shows more of the semantic-based +Interp Φ -features (paralleling to a certain degree what one might expect of VP/Thematic relations, under the postulation that Interpretable features are semantic in scope and not part of the formal syntactic numeration). Hence, null subjects of the Italian, Spanish variety are inextricably linked to the rich morphologically agreement system.

- [44] It is therefore not too unreasonable to assume that aspects of AGR / INFL in pro-drop languages may sustain a certain amount of subject incorporation (on a scale with subject-verb grammaticalization). Such inherent inflections could be considered as having semi-formulaic tendencies, or rather could be viewed as taking on characteristics of incorporation, similar to how the derived two-morpheme word *teach-er* was said to be processed as an inherent chunk. English children never delete the derivational affix {-er} even when many such true inflectional affixes are being dropped all around them. English children omit the infinitive 'to' inflection early-on in stage-1 of their production precisely because it constitutes a 'true affix', unincorporated marker (as opposed to other early affix markers such as participle forms {en} and {ing} which seem to behave as non-affixes).⁸ Such participle forms maintain 'strong-stem-like' properties and thus may share some manner of incorporation. What is interesting about early Italian data is the fact that AGR inflectional errors are almost non-existent (a hallmark of rote-learned/lexical productions) despite the fact that other affix deletions do manifest at the same stage in question. For instance, what one typically finds at the early stages of Italian syntactic development are (*inter alia*) errors regarding Number on Noun stems (a possible syntactic bare stem production).
- [45] What this might suggest is that when the early child data are examined carefully, distinctions regarding the type of inflection can be made, leading to discussion of whether or not a production should be considered as a true stem+affix inflectional production. Italian here may constitute a periphery case, but when taken in conjunction with more agglutinative type languages, the findings support the general idea that inflections for [-bare stem] languages may to varying degrees be incorporating the affix into the stem. If an inflectional affix is routed and incorporated into the stem, than processing access of the affix has the same status as access of the stem—with a kind of lexical incorporation that mimics pie-piping strategies. It would seem that non-clitic, true rule formations of inflectional morphology would not trigger such strategies. In brief, developmental studies show that when one has to use inflection on every form [-Bare-stem parameter],

children at early syntactic stages of development seem to project only one inflected form for each word and only produce that inflected form (as the default). This default is presumed to be based on frequency. For instance, it has been reported that children may (i) mark the most salient affix, say number on nouns, (ii) but then still confuse how to properly project number on each noun type for e.g., gender and case. It is suggested herein that the former affix process (i) should be thought of as a kind of lexical incorporation, showing no true rule process, with the latter paradigm process (ii) being viewed as constituting a true rule-driven process.

fMRI, Neuro-imaging, and the Dual Mechanism Model

- [46] In relatively short order, great strides have been made in securing a working mapping of the brain, as it has to do with a subject's partitioning of language. Results obtained in studies using data from fMRI (functioning Magnetic Resonance Imaging) as well as ERP (Event Related Potential) have now been compiled and analyzed from various neuro-imaging labs across the world. These data are bound to make an increasingly important impact on the debate now being waged regarding the connectionists' visions of a single mechanism model versus that of nativists' visions for a rule-based dual mechanism model.
- [47] It seems well established now that language triggers patterns of frontal lobe activity (Broca's area associated with syntax-based tasks), temporal lobe activity (Wernicke's area associated with lexical retrieval tasks), as well as other left dominated language areas such as the cingulate gyrus and the dorsolateral prefrontal cortex. The overall data collected thus far seem to largely support the traditional view held that language is—(i) left hemisphere dominated, and is (ii) partitioned in two main ways which speak to the classic distinction between semantics vs. syntax (as suggested by proponents who theorize a Dual Mechanism Model (DMM), or, minimally, that language is (iii) partitioned by semantic vs. phonological procedures (as suggested by those seeking a single mechanism Connectionist Model) (CM). What may turn out to be most important, however, for the views expressed herein, is that it exceedingly looks more and more likely that a convergence is taking hold amongst both models whereby assumptions granting language the status of being biologically determined is the null hypothesis—hence opening discussion into maturational consideration having to do with specific lateralization. (See Elman & Newport for such remarks on CM, Marcus on DMM).
- [48] Let's regard the front on brain mapping. It continues to be true that the brain affords very little to the investigator in the way of any simple and direct 'language-to-mapping' correlation—that is, outside of the basic notion that language is dominated by the left hemisphere, coupled with what had been known for quite some time regarding Broca and Wernicke language control areas, very little remains understood in actual substantive measure regarding brain-to-language processing. Despite the stubborn heterogeneous nature of the brain, a consensus is brewing nonetheless in neuro/psycho-linguistic circles suggesting we are just now starting to view a more comprehensive picture of the subtle brain

- engaged in language. For instance, fMRI researchers are now learning how to control and isolate the language task by asking subjects to ‘think’ of words rather than to ‘speak’ them, thus controlling any motor-artifact that would be associated with speech, etc. These and other concepts, along with their implementation in design, have taken researchers one step further in our ultimate understanding. Prior, much of the data of any one language task was essentially contaminated by what we called extraneous processing noise. However, given that such strives have been made, most of the empirical data thus far collected on the brain, at least as it has to do with the specific question of the DMM, have been fuzzy at best.
- [49] For instance, one would have hoped for a clear and distinct fMRI picture to emerge with regards to say INFlectional vs. non-inflectional aspects of morphology. But none is yet on offer. For example, in some studies, regular and irregulars fail to provide a crisp distinction regarding brain imaging: while some regular verbs (e.g., *walk-walked*) take on irregular priming effects (Alegre & Gordon, 1999; Clahsen 200-) other irregular mixed verbs (e.g., *feel-felt*, *wife-wives*) take on regular priming effects (Marslen-Wilson, Hare & Older, 1993). (See Grodzinsky who suggests that only certain aspects of rule-driven grammar such as movement operations apply to Broca’s area processing. Steven Pinker and associates at Harvard are much more up-beat about potential clear-cut distinctions). One common factor that may play a role in this INFL-priming continuum has to do with the verb’s frequency. It may be the case that a high-frequency regular verb such as *walk* (typically associated with the rule procedure assigned to Broca’s area) rather begins to cross-over onto a different processing plane whereby rules get supplanted by more robust associative procedures (Wernicke’s area). Such a potential cross-over effect would have an explanation consistent with the DMM framework. We will keep some of these and other questions close to us as we begin to probe further into the theory of the DMM.

Remarks on Borer: ‘Minding the Absent’

- [50] The language input therefore must somehow first be tagged (*a priori*) prior to its entrance into the speaker’s language modular of the brain. In other words, processing itself doesn’t act on the language, but rather the language itself, removed from linguistic processing, is self contained. Language defined this way would be self generating and reside, in the first instance, outside of the brain/mind of the speaker. Recall, J. Fodor’s notion of ‘treelets’ suggests that language development/parameterization comes to the child in small pieces of tree structures already composed (with parsing material encapsulated), and that the child doesn’t really engage in any pure bottom-up parsing of language in the traditional sense of the term. I am not quite sure how we are to understand such an outcome short of saying that e.g., when Chomsky speaks of language as being a ‘real object’, we are to take that as meaning a pure (objective) object whether or not a (subjective) speaker comes into the equation. Chomsky may in fact subscribe to such a view, particularly when the idea is framed around public vs. private language. But to say that language creates itself outside of the brain/mind is altogether another proposal.

- [51] All this speculation seems to center on one leading point: viz., the fact that certain elements of language go missing from the early onset of language suggest, above all else, that in order for something to go missing, it must already have been potentially selected: hence a kind of *a priori* tagging of language. In order for ‘bits of language to go missing’ (i.e., lacking processing), as Borer suggests, there must be some mechanism that first partitions the language into categories prior to the phonological/syntactic interface. More specifically to this idea, a viable maturational hypothesis—as seen through the eyes of both Chomskyan modular research and brain mapping research—has come to consider that linguistic categories must somehow subserve different neural substrates. The question then arises as to how we are to understand this *linking* between the language component and the neural component—viz., does the brain self regulate and categorize the language input (top-down), or conversely, do language particulars somehow seek out and assign appropriate neural substrates (bottom-up)?

Lateralization and Maturation

- [52] It is my feeling that the remarks of B&R here can be used to leverage the larger question at hand—namely, as to whether or not maturation affects lateralization, which in turn may provide a biological basis in accounting for the typical syntactic stages of child language acquisition called for by B&R. For instance, if young children between six and eight years of age show a marked increase of lateralization in Broca’s area (that region claimed to support rule-based procedural language/syntax), then B&R’s remarks for an earlier ‘non-functional’ stage-1 as based on tacit ‘*Knowing-causa-deletion*’ (rather than my presumption of ‘*knowing-causa-no access*’) might, in general theoretical ways, become somewhat questionable. To be more concrete, my own claims herein will be organized on the premise that while I do agree with B&R on the observable data (i.e., their claim for a non-functional stage in children’s syntax, see Galasso 1999, 2004a/b), I believe their notion behind such a ‘*knowing-causa-deletion hypothesis*’ is, at the very best, misleading. Recall, as B&R point out, that whatever the case may be between the two notions of ‘knowing’ here, it remains incumbent upon ourselves to offer some account for the child’s exclusive omission (and not commission) of functional categories. The term ‘knowing-causa-deletion’ seems to me to fall short of offering any complete account of omission and basic questions quickly surface: viz., in what way does ‘knowing’ result in ‘deletion’, particularly when we are talking about a tacit knowledge in the sense of operating on a set of parameters determined by the Language Faculty? I’ll attempt to show below that arguments which surround the DMM may also parallel Borer’s observations and extend them in ways which speak to a ‘language to brain-maturation correlate’.
- [53] General theories linking the incremental myelination of Broca’s area to the eventual onset of functional categorization have sparked interest into what Borer’s observes to be this clean-cut asymmetry between what is lexical (phonologico-semantic level) and what is functional (syntactic level). Wilcox et al. (199-) make

explicit this point in their paper

Specific brain imaging research investigating the role age has in acquiring syntactic complexity has been in relatively short supply. Only scant research to date has been undertaken in ways in which very young normal children (between the ages of two and five) are fitting with fMRIs, and where module-specific language tasks are designed to see just how and where in the brain the actual acquisition of languages takes place. For example, Holland *et al.* (2001) were one of the first fMRI studies to address this question of whether maturational (age) was involved with Broca's lateralization. The findings suggested that when age was factored, there was not only a linear increase of lateralization (mostly seen in left hemisphere/Broca's area), but also that the actual month-to-month, year-to-year increase could be attributed to the number of newly acquired lexical items the child was able to retrieve. In other words, notwithstanding a multitude of different factors which could go into explaining this lateralization, chief among them being the notion of maturation, Holland *et al.* suggest that the mere number of lexical items the child had to store and retrieve over time might be a factor in this shift and increase of right to left hemispheric lateralization. I return to this notion of 'memory spawning rule-based syntactic processing' later on in this paper. Meno's paradox and the bootstrapping dilemma

- [54] The great epistemological problem known as *Meno's paradox*(Plato) reveals to us that we cannot (*inter alia*) just roughly pose and entertain a question to any given problem without first knowing the proper answer to the question. It seems to my mind that Meno's problem becomes very relevant when we begin to closely exam language acquisition, as made apparent by Borer's very own ideas of language knowledge and learning—a dilemma referred to by developmental linguists as the bootstrapping problem. The bootstrapping problem tends to come in two forms: (i) the 'associative' driven *phonological/semantic bootstrap* hypothesis (see Jusczyk & Derrah (1987) and Morgan & Demuth 1996 for a review) as it opposes (ii) the 'rule-based' *syntax bootstrap* hypothesis (see Gleitman, 1990; Naigles 1990, 1995). The question at large I pose in the following subsections attempts to link Borer's remarks on functional categorical deletion to both bootstrapping hypotheses, as coupled with what we know regarding the emerging DMM.

Associative Processing: lexical-phonologico-semantic

- [55] Our aforementioned interpretations on Borer lead us to believe that by 'deletion', we must rather mean 'non-access'. The very observations that Borer draws on regarding her '*knowing-causa-deletion hypothesis*' can be reinterpreted to fit the so-called associative bootstrapping model, particularly as it has to do with phonological/semantic-based word learning. Notice that both phonology and semantics are the two preferable processes of choice by which connectionists' models are implemented. Research using this approach shows that very young children in fact do 'know' (using Borer's terminology) and can perceive both lexical and functional categorical language. The evidence for this knowledge has come out of designed experimentation which seeks out to determine if young children know e.g., that a functor/functional word must be present within a phrase

- boundary. For instance, in such ‘head-turn’ experimentation, very young children react differently to Determiner phrases which provide the appropriate syntactic Determiner marker at the phonological level, such as a schwa insert (e.g., ə *book is there*) as opposed to there not being one (e.g., Ø *book is there*). The general findings suggest that children begin to acquire and pay favoritism to the sound sequences which form word and syntactic boundaries consistent with the phonotactics of their target language (see Thiessen & Saffran 2003).
- [56] It indeed may be the case that young children in these experiments do in fact ‘know’, tacitly, that the word class *determiner* belongs and must insert before the word class *noun*. As Borer shows, the very fact that children exclusively ‘delete’ only functional words—whereas children don’t randomly and unsystematically treat both lexical and functional categories with omission and commission, a kind of deletion which might otherwise point to performance glitches of production rather than to grammatical/categorical distinction—surely suggests that the children know the difference between the two word classes.
- [57] Where I fundamentally differ with Borer is that I believe the grounds for this surface knowledge can readily be found on the basis of (associative) phonologico-semantic distinction, rather than on the exclusive (rule-based) knowledge of the syntactic class *per se*. (We can readily assume that children have heard, at a very high frequency, the sound pattern associated with a DP as it embodies a phonologically salient pre-Noun determiner (schwa) then preceded by a given syntactically salient Noun). Having accepted this, the question still remains how and why it is that children ‘know’ the functional word class, yet fail to produce them at the relevant stage-1 of speech? Recall Borer’s claim—as it jibes with a number of developmental researches proposing such a non-functional stage—that catastrophic functional morpheme/word deletion pertaining to inflectional morphology and functional word class (respectively) takes place in the developing mind of a child, and this non random deletion provides *ipso facto* evidence toward knowledge of word class distinction. The idea that I would like to propose here ties together the DMM with notions of speech production as it has to do with syntactic saliency of certain word categories.
- [58] **Retuning to general questions of theory**
Returning then to general questions of theory, Chomsky (1999) suggests that there is good reason to posit a set of universal features along with principles that make up the Language Faculty. The problem facing the child may be how to build-up (bottom-up) such universal features into a lexicon (see fn. 8). If indeed a bottom-up assumption underpinning the processes of lexical construction is correct (i.e., that the knowledge of morpho-phonology precedes and triggers syntax, much in the spirit of Distributional Morphology, or the Lexical Learning Hypothesis), then there is no reason to suppose that the assembling of features does not follow an

incremental process, potentially governed by specific maturation of cognitive complexity. In other words, some features might be acquired before others based on their internal conceptual complexity. For instance, we might hypothesize that the internal complexity of the features [PER(son)] might be more complex than the feature [DEF(initeness)] since DEF may contain some amount of cognitive relevance at LF). (This range of complexity can have large consequences for overt checking. For example, under Guasti and Rizzi's (2001) model of spell-out, it is assumed that differences between overt vs. covert checking directly corresponds to where in the position of the syntactic tree a phrase projects. Those lower positions which are dominated by the VP may include only +Interpretable features, and so may survive without forcing overt movement to a higher projection. Higher positions dominated by IP conversely force such overt movement for the required checking of functional -Interpretable features.)¹⁰ Hence, the agreement of PER as well as the agreement of structural CA might burden an otherwise primitive computational capacity for a young child at our stage-1 of language development. However, this can't be the whole story since some language groups manifest very early functional categories/features in child development. It seems the cognitive complexity of features must work in tandem with the broader implications of morpho-syntactic typology subsumed under the matrix [+/-INFL]=>[+/-Bare Verb stem]. More concretely, it goes without saying that not many linguists would wish to posit features as containing different saliency values from one language to another. Formal features are part and parcel of a universal language faculty, a C(omputation) built-up on the H(uman) L(anguage) cognitive template (sometimes labeled as C_{HL}). Hence, all properties of feature complexity must be in the first instance universal. It is then in the second instance, where formal features overlap onto [+/-INFL] language typology, that we can ultimately determine whether or not the given language will exhibit early functional projections in child language.

- [59] As a follow up to §§57 and 58 above, one of the many things that remain a puzzle to my mind is the following question: Why is it that early child English allows for [-FIN] bare verb stems to occur in Wh-Questions (*What he eat?*)? This structure has only been reported for Child English and, I am aware, has never been attested in other cross-linguistic child studies. It begs the question: What makes the English bare stem so special in this respect, how does it *differ* to other language infinitives (e.g., root infinitives), and how does it function in the syntax? Of course, one way to react to such anomalies is to simply reconsider it as a [+FIN] bare verb stem, with only the phonological affix missing.¹¹ If this is at all possible, then it begs a larger question: Can functional material be stored and represented in two fundamentally different ways? It would seem language typology demands it. Surely, the PF-overt vs. LF-covert movement analogy captures actually this. So, we may go on to assume that such an 'affix-less' verb nevertheless projects inflectional material (T/AGR), at least in light of the fact that there is nominative subject case agreement (*He*). One means to create a dialogue about the *difference* is to set up a heuristic procedure. The idea being if we can somehow capture the properties behind English bare stems, we might be

better able to understand the overall syntax of (Child) English and throw some light onto how English morphology fundamentally differs to other language groups, at least in this one small respect.

- [60] The two-point heuristic procedure takes the following shape (inquiring into continuity between the child and adult grammars):

Question #1: Complexity of Features:

Q: Is there Continuity of Features?

- If yes, see question #2. (We take this to mean that T/AGR features project from the very earliest MLU).
- If no, then a Discontinuity model is appropriated.*
 * (Since English is a [+Bare Stem] Language, it can simply exhibit bare verbs stripped of its formal features. Nothing hinges on the fact that that it exists within a Wh-question (an otherwise functional projection). What then must be looked at is the pending nature of the Wh-Question (noting the absence of an Auxiliary) not the bare verb, as Wh-operators equally can be expressed at PF without exhibiting the complete range of formal features attributive to CP. I understand the assessment fails to deal with optional projections).

- [61] Typically, the notion of +/-Interp(etable) features comes into play in helping to account for the asymmetric chronological onsets of the relevant features and functional projections. It is suggested that [+Interp] features provide semantic material at LF interface while [-Interp] features must be functionally checked and erased at LF (Chomsky 1995). In line with the arguments in this paper, it could also be suggested that the two types of features likewise get processed and stored in two different regions of the brain—whereby [+Interp] takes on ‘strong-stem’ properties and [-Interp] takes on ‘weak-stem’ properties (See §24 below regarding +/-Interp features in child language acquisition.)

- [62] Question #2: Complexity of Paradigm:

Q: Is there Continuity of the Inflectional Paradigm?

- If yes, morpho-syntactic typology must be considered, triggering the Dual Mechanism Model as a possible means of accounting for the continuity. (We take it here that some amount of feature projection may in all actuality be governed by phonological constraints on the language, such as cliticization, lexical incorporation, etc.)
- If no, Then a discontinuity model is appropriated.

In sum, we take it that Discontinuity is the Null Hypothesis, and that any evidence to the contrary must be reconsidered in relation to that language’s morpho-syntactic typology. The exact role of the typology here will be made clear below.

[63] **Some Data**

Evidence for a Discontinuity model is striking ('negative' to Question #1 §18). For instance, Radford and Galasso (1998), Galasso (1999, 2003b) Radford (1999, 2000) provide English data showing that children enter into a 'No Agreement' / 'No Inflection' initial stage-one of acquisition during which they *completely omit functional categories and [-Interp] complex features*.

[64] Stage-1: 'No AGREement-No INFLection' (Radford & Galasso 1998)

Possessives: That Mommy car. Me dolly. No baby bike. Him name.
Have me shoe. **Iwant* me bottle. It me.
Question: Where Daddy car? This you pen? What him doing?
Declarative: Baby have bottle. Car go. Me wet. Me playing. Him dead

*(*Iwant* examples are analyzed as formulaic chunking, since no other supportive material providing for a functional analysis of nominative case is found in the relevant stage).

[65] Stage-2: 'OPtional AGREement -INFLection'*

Possessives: That's Mommy's car. My dolly. Baby's bike. His name.
Question: Where's Daddy's car? This is your pen? What (is) he doing?
Declarative: Baby has bottle. Car goes. I'm wet. I'm playing. He's dead.

*(The OI stage (as suggested by Wexler 1994) would simultaneously incorporate both data sets as described in his initial Optional Infinitive stage-1). Radford & Galasso make a clear demarcation between the two stages, with the complete absence of any optional functional projections for their stage-1. For complete data/analyses, see Galasso 2003c).

[66] Overall, children in this initial stage-one of syntactic development are forced into projecting very limited structure. For instance, (and this is by no means an exhaustive list):

- (i) Possessive projections, which rely on an AGREement relation with a nominal INFL, must default to an objective case (e.g. *my* to *me*);
- (ii) Verb projections are limited to VPs without INFLection (hence auxiliary-less question and declarative bare verb stems) (e.g. *What him doing?*, *Car go.*);
- (iii) Subjects, which rely on an AGREement with a verbal INFL, must default to having an objective case (e.g., *Me wet*). Consider the syntactic structures below pairing the two data sets, with stage-one showing no inflectional phrase (IP) agreement.

[67]	<i>Structure: Stage-One / -AGR</i>	<i>Structure: Stage-Two /+AGR</i>
(i) Possessive: *	[IP Mummy [I {-agr}-ø] car]	[IP Mummy [I {+agr}'s] car]
	[IP Me [I {-agr}] dolly]	[IP My [I {+agr}] dolly]
(ii) Case:	[IP Him [I {-agr}] dead]	[IP He [I {+agr}'s] dead]
	[IP Me [I {-agr}] wet]	[IP I [I {+agr}'m] wet]
(iii) Verb:	[IP Baby [I {-agr} have]]...	[IP Baby [I {+agr}has]]...
	[IP Car [I {-agr} go -ø]]	[IP Car [I {+agr} go-es]]

*(Though Radford and Galasso do label their stage-one structures as IP {-agr} in the 1998 paper, this is mostly notational: see *in-note* in §68 below).

- [68] Similar findings are born out and widely attested in the literature and are consistent with the general notion that language acquisition involves some sort of *incremental feature-building* (Radford 2000)—viz., the notion that if language does proceed in an incremental way, then it should be of little surprise that the more robust and primitive aspects of a language should come on-line before more abstract aspects of language—specifically, default {-agr} feature projections attributed to the VP (by default) come on-line before {+agr} projections attributed to the IP. (*In-note:* Though Radford and Galasso do label their stage-one structures as IP {-agr} in the 1998 paper, this is mostly notational: they are clear in stating that this in no way should undermine the fact that we are indeed dealing with a lexical stage-one, with IP remaining Non-specified {Ø} (and for all intents and purposes, not projecting). Galasso (2003) describes such early stage-one structures as reduced to simple lexical VPs and NPs). This gives us the flavor of saying that a *maturational scheduling* is behind the chronological ordering of features (much in the spirit of the Brown studies (1973) which sought to show a time-line of affix morpheme development—moving from potentially viable semantic-based participle forms {en}, {ing} through to true rule-based inflectional forms 3PSg {s}, Possessive {s}, Past Tense {ed}. Where the opinions of the cited authors §§1,3 tend to diverge, however, is in how to appropriately and accurately describe such apparent lack of child linguistic knowledge. Given this divergence, arguments spring up as to whether or not there is really any claim for discontinuity at all (as based on the research paradigm on offer (Chomsky 1995, 1999)). One might think that the very fact that the child doesn't produce/generate target 'adult-like' grammar at the initial stage of syntax should be enough in of itself to say that discontinuity is an accurate depiction of child language development.

- [69] Well, some of us take maturation more seriously than others. For the sake of concreteness, let's spell this out. The 'strong maturational' theorist believes that there is indisputable evidence for some kind of a biologically determined stage during which functional categories, along with their class of functional words, parameter settings and formal [-Interp] features, are all together absent in the child's grammar. In these terms, 'strong maturation' equates to 'discontinuity'. The 'weak maturational' theorist might be more prone to skirt such biological issues and stake their arguments around interpretations of the theory. For instance, one way around the strong maturational/discontinuity account would be to say that all formal categories/features are available to the child from the very beginning.¹² The apparent discontinuity is then reduced to the mere phonological spell-out of yet un-set parameter settings of categories and/or un-specification of features.
- [70] Hence, there are two schools-of-thought of how such omissions of structure should be explained:

- | | |
|-------------------------|--------------------------------------|
| (i) Full Competence | => (weak maturation-continuity) |
| (ii) Gradual Competence | => (strong maturation-discontinuity) |

Full Competency in this respect would include any developmental model which credits the very young child with at least knowing the functional projection notwithstanding the fact that the features of the projections can go unspecified or be phonologically null (cf. Wexler's initial OI-stage 1994).

Gradual Competency in this respect would include any developmental model which holds off crediting the very young child with knowing any functional projection that doesn't at least make itself phonological present in the form of functional categorical words (Case/Pronouns, Auxiliaries) (cf. Radford & Galasso's initial No-INFL stage 1998).

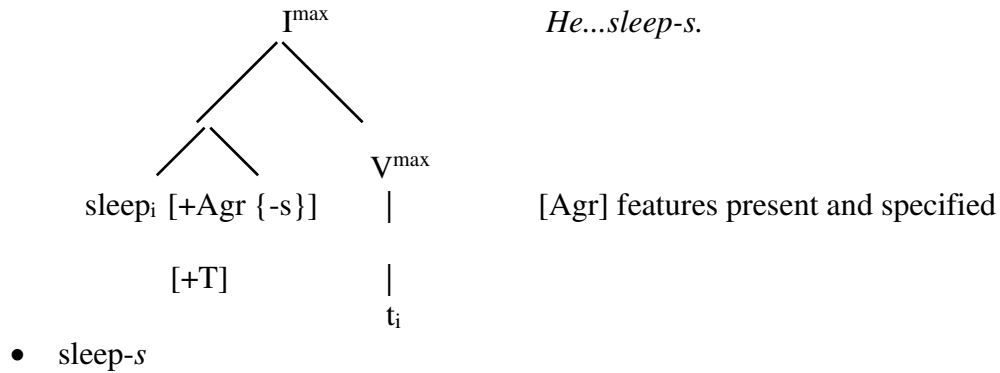
- [71] Full competence hypotheses state that all functional categories/features are present in the child's linguistic system (this equates to a child-to-adult continuity). The child's otherwise non-target grammar is thus the result of un-specification of features that are nonetheless present in the child's phrase structure. In other words, the child's otherwise lexical non-functional stage-1 of grammatical development is thought of as maintaining the full-fledge IP>VP phrase structure for typical declarative sentences, non-specification of features notwithstanding, hence, the early German and French data cited above. In English however, the child may not spell-out the Phonological Form of articulation (at PF) of such features—for instance, the dual AGR(ement)/T(ense) feature (which triggers the inflectional affix {-s} for 3person/singular/present) on the main finite verb goes un-specified and therefore doesn't project. In this sense, a non-inflected bare verb-stem surfaces as the default: e.g., *(The) baby sleep*. What this assumption basically claims is that a child may in fact be producing functional (affix) material all the while despite the fact that such material is phonologically silent (following assumptions laid out in Halle and Marantz 1993). Recall, such a treatment was suggested regarding Bare verb stems which show subject nominative case in

embedded Wh-Questions §17. This very real possibility that functional categories/features can be present though phonologically silent changes the entire landscape of how maturational theories of language acquisition can be put to the test. The very idea that +affix material can be *present* in a grammar without actually being *spelled-out* in the phonology requires developmental linguists to revisit their discontinuity theories to the extent that newer models need to be devised in order to establish whether or not a child at the earliest MLU stage of development has functional grammar.

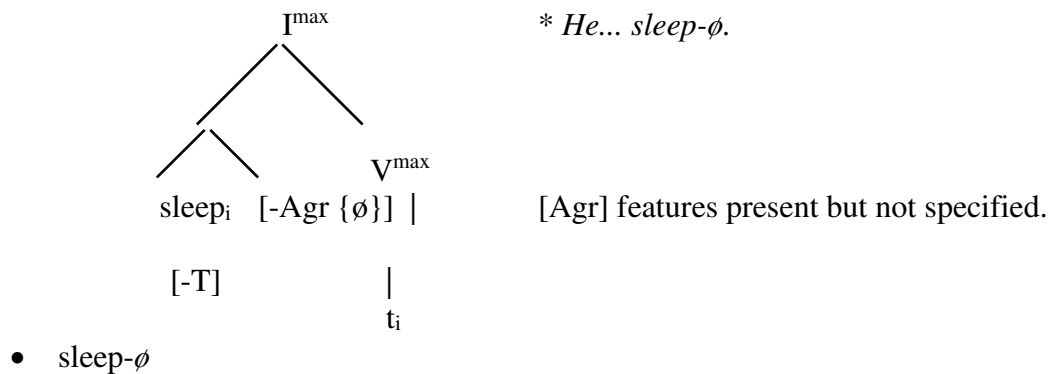
[72] **A new condition for a non-functional stage-1**

The argument for potential underspecification of features can be naturally expanded to suggest that there are indeed two ways in which the brain can symbolically represent morpho-syntactic material (The DMM)—either via an overt PF channeling, or via a covert LF channeling. This will be later discussed in the following sections on typology. In any event, what I am on about here is that the two approaches (continuity vs. discontinuity) can converge to express a larger idea: namely, children may have an alternative option of symbolically representing functional material, but that this option is only relevant to the +/-specification of affix material in general. In other words, we have now conditioned any strong maturational account and relegated it to a stage that describes the complete absence of functional material, but only insofar as it is typically expressed in whole words (such as the Pronouns (case) AUXiliaries (movement), etc.), and not affix and/or clitic constructs. Given a child now has a parameter option of projection affix material in two ways, any attempt to define a non-functional stage solely based on the absence of affix material without looking into other paradigms has become considerably weakened. (If anything, the option speaks to a *pre-parameterization* stage, which, by the way, would also support a strong maturational hypothesis). Having leveled this new condition for our non-functional stage-one, it must be said that Radford & Galasso's Lexical stage-1 remains upheld, albeit now not by the fact that affix inflections are absent (as their absence at PF/performance may not provide evidence for the absence at LF/competence), but rather the no-functional stage remains upheld rather due to the lack of other functional material such as CAse and AUX. as attested in the data. Consider the two analyses of the verb's inflectional material below (with §73 showing a common analysis).

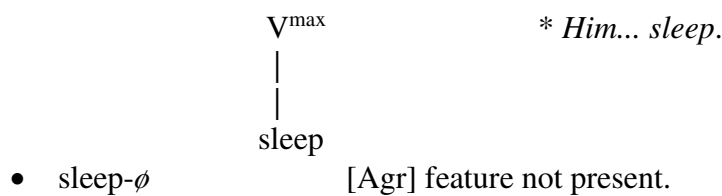
[73] *Full Competence Phrase: Features spell-out for target grammar*



[74] *Full Competence Phrase: Features are present but un-specified:
(Bare verb stem by default)*



[75] *Gradual Competence Phrase: Features not present
(Bare verb stem by default)*



* (The trees above show AGR working in tandem with T. It is in this sense that the new condition as expressed in [§30] holds regarding affixes (in this case, the Agr/T-affix). It is rather the relation of AGR/CAsE that now becomes central in

- identifying a no-functional stage. Recall our data in §21: there are no attested data of productive Nominative Case. It is assumed under Schütze & Wexler's (1996) model that the T/AGR work together, though see Galasso 1999 (p. 94-5) for a different treatment showing the affix {-s} exclusively to mark Tense, whereas the full agreement paradigm throughout goes unmarked [-ø] due to the nature of an invisible agreement assigning mechanism in English)
- [76] Regarding the Gradual Development Hypothesis, the main idea behind the child's deficient grammar is based on the complete absence of formal [Agr] / [-Interp] features and the subsequent lack of functional categories IP/(CP). [Agr] / [-Interp] features are those formal features such as C_{ase}, PERson, Tense and other AGRreement relations which are considered to contribute no substantive semantic material toward the overall computational system (the locus of Logical Form).

[77] **Conclusion: Inflections and Typology**

'Modular-complex' (English, Italian) vs. 'Modular-simple' (Chinese, Hungarian)

Let's turn to our closing remarking on typology. Languages differ in whether or not they are *analytic*, *agglutinative*, or *synthetic*. A finer-grained treatment of this classic typology, however, shows that while the first two types maintain a rather modular-simple representation of functional, inflectional material, the latter may show a modular-complex representation. In this sense, inflection is tied to typology. In this final section, we make explicit what we mean by this *modular parameter*.

- [78] In addition to the classic typology stated above, past typological studies have shown (e.g., Hawkins, 1986; Comrie, 1987) that the three language types actually can collapse under a dual-typology based on semantic vs. syntactic/grammatical structure of the language. More concretely, a scalar model showing a continuum from syntactic to semantic structure can be devised to help define the way a given language works as a communicative system. The continuum demonstrates the competing elements a language faces—namely, how does a language balance the needs to communicate *effectively* (semantic-based) with the needs to communicate *efficiently* (syntactic/grammatical-based).

<i>Modular-Simple</i>			<i>Modular-complex</i>		
Ø INFL	Chinese	Hungarian	Hebrew	English	Italian Latin
-/+ INFL					
{----- -----}			{----- -----}		
effective/semantic			syntactic/efficient		
[+ Lexical Incorporation]			[- Lexical incorporation]		

The former deals with semantic properties of a language since it involves the content of speech (*per se*), while the latter involves crucial issues of syntax, since it is syntax that allows the language system to evolve from *complex* to *simple*, a move in keeping with all biological systems: (the need to economize).

[79] **The Proposal**

We believe this dual-typology of language, as based upon *transparency of semantics vs. syntax* (termed modular-simple vs. modular-complex), more-or-less reflects the two fundamental ways in which the human brain is constrained in the structuring, storing and projecting of linguistic material (The Dual Mechanism Model). Similar characteristics of this proposal have been advanced in the literature (regarding verb movement type) which suggests that overt subject languages (such as English, French) use ‘adjoined-affix’ morphology as opposed to ‘head/stem-affix’ morphology as seen in covert subject languages (such as Spanish, Italian).

We take it that semantic transparent (modular-simple) language types, which seek to preserve the optimal level of meaning in language, ultimately employ more of the temporal-lobe regions of the brain in constructing their grammars. Modular-simple language types thus organize grammatical relations as defined by thematic/semantic roles. This leads such languages toward two extreme ends of the continuum—either they show pervasive inflectional marking (based on a strict one-to-one-morpho-syntactic mapping onto semantic roles), or they go without inflection all together. The semantic continuum thus looks like the following:

[80] *The Semantic Continuum* (modular-simple):

- (i) *Extreme Markers*. This language system is designed around extremely high levels of lexically embedded morpho-syntactic markers, dealing with a wide range of semantic/grammatical relations—e.g., topic, case, temporal, locative, variable word order, etc.—*agglutinative* languages follow this course of design...

....or,

- (ii) *No Markers*. This language system is designed around an almost complete absence of such markers—*analytic* languages like Mandarin Chinese. Both language types can be similarly characterized by their high levels of lexical incorporation of such markers—‘a temporal-lobe design’.
- Conversely, we take it that syntactic and thus grammaticization-based languages seek to preserve efficiency in ways that collapse semantic roles in a variety of way (prone to complexity). (This ‘complex-to-simple’ continuum can be seen in diachronic language change, respectively, from Latin to French, from Sanskrit to Hindi, from Old English to Modern English, etc.). The syntactic continuum looks like the following:

[81] *The Syntactic Continuum* (Modular-complex)

- (i) *Rich Inflection*. This language system is designed around a stronger paradigm of inflections, though such inflections are minimized and semantically

deflated since they no longer have real antecedents to semantic roles—*inflectional* languages like Spanish, Italian...

...or,

- (ii) *Impoverished Inflection.* This language system is design around a weaker paradigm of inflections—*inflectional* languages like English, French.

[82] *Modular-Complex.* It may be that English children initially assume their target language to be simple, or at least paradigm consistent (the default assumption). Accordingly, children in their initial stages of syntactic development, may wrongly assume (and thus misanalyze) that many of the ‘slight’ imperfections in their language are in fact perfections. In other words, children may be perfect language learners *in the sense that they are perfectionist who seek perfection in the imperfect input they receive* (Radford 2000). This brings up one aspect of complexity, the competing nature of syntax over semantics. The fact that language acquisition involves the incremental construction of features from simple to complex (from +Interpretable/semantic to -Interpretable/syntactic), does suggests that such mapping is correlated to (maturational) brain development. The incremental nature of syntactic development begins with the VP/Thematic structure (reported to belonging to the more primitive, temporal-lobe region of the brain), and advances outward to the formal phrase projections of IP/CP (belonging to the frontal-lobe).

[83] Consider the fact that, initially, young English children wrongly analyze irregular verbs as (non-past) bare verb stems. Hence, such over-regularizations as *wented*, *boughted*, *thoughted* should not be considered as double past markings. The fact that the early treatment of irregulars doesn’t encode for tense suggest that the child first stores the irregular verb as a non-modular strong-stem. (Hence, inflectional over-regularization in Early Child English (Marcus et al. 1992) may be the result of a child wrongly assuming a ‘modular-simple’ setting over a ‘complex’ setting.) Young English children generate a number of such syntactic/inflectional errors early on in their development, all of which point to this mis-guided parameter setting: e.g., double tense markers as in Aux constructions *I didn’t saw/had/played the toy*, suggesting that the Aux lacks a tense marker, Case, Agreement and Tense deletions as in *Him do it* may be due to weak relevant paradigms, etc. Also reported above, children may optional project either way and thus mix the two settings, as when they start out speaking bare verb stems in otherwise functional projections (*What he eat?*). These examples, when taken as a whole, demonstrate, two main points: (i) the learning burden imposed by such imperfect irregular formations, and (ii) that the processing of paradigmatic functional material may be stored and projected in alternative ways (depending on the cognitive maturational schedule of language development). This duality is what is ultimately responsible for a two-stage Lexical vs. Functional development of grammar. More specifically, the claim here suggests that when proper functional material projects in the way of an inflectional affix, it does so by way of a storage capacity that treats stems and affixes in an entirely different manner, as the two elements, stem+affix, are stored in entirely different

parts of the brain. When the child's brain is not yet prepared to handle such (frontal-lobe) activity, the linguistic formation may fall back on the rather default handling of material indicative of more semantically-based structuring (temporal-lobe). The proposal is that proper affixes are stored in the frontal-lobe regions of the brain, unconnected to the stem, which is itself stored in the temporal, motor-strip regions of the brain. Modular-complex languages, however, must also adhere to an additional parameter which determines whether or not the language is richly inflected (and thus [-Bare-stem], or weakly inflected (and thus +Bare-stem). It is in fact this wide range of detachment of stem and inflectional affix that procures the computational processes as handled by the frontal lobe, otherwise, as with the other two typologies, a single processing mechanism correlating to the temporal-lobe suffices. When these two parameters are taken into account, a picture emerges as to how the development of child functional grammar should manifest:

- [84] (i) *Modular-complex / -INFL*. These languages (English, French) should demonstrate a initial stage-one of syntactic development where functional inflections go un-projected, as attested in English/French child language.
- (ii) *Modular-complex / +INFL*. These languages (e.g., Spanish, Italian) should demonstrate an initial stage-one of syntactic development where functional inflections project—at least partially and at times presumably incorrect.
- (iii) *Modular-simple / +INFL*. These languages are the agglutinative types with highly regular inflections (Hungarian, Turkish, Russian, Japanese). The inflections are not only highly regular but also correlate to a one-to-one correspondence in meaning (one morpheme, one meaning). Consider the Turkish word *odalarimdan* ('from my rooms') where *lar -im -dan* are morphemes expressing one distinct category. Such corresponding could be similarly viewed as Derivational morphology in a certain respect since each added morpheme can change the stem meaning. Young children speaking these languages do not typically pass through a non-inflectional stage, since the inflections are housed and encoded within the stem. In other words, young children speaking Russian or Hungarian don't seem to follow the traditional benchmarks or morphological development as compared to their synthetic language counterparts. (A good illustration is the case of Japanese post-positions and locative markers which tend to heavily rely on the substantive/semantic properties related to the stem. Such markers cannot go deleted in early Japanese syntax, unlike other non-substantive Japanese INFL markers which do (nominative/accusative Case *-wa*, *-ga*, *-o*.)
- (iv) *Modular-simple / -INFL*. These languages are the pure analytical languages such as Mandarin Chinese which tend to 'unpack' the little inflectional morphology they have and established otherwise potential morphological markers as part and parcel of the stem—e.g., the English paradigm of First Person, Subject, Plural which is 'morphologically packed' in the pronoun *we* (and subsumes all markers) would be expressed in Mandarin as three word boundaries: #[=one person] + #[=many of].

[85] **Brain-to-language processing and a maturational story for modular complex languages.**

Regarding the brain, I feel that it eventually won't take much to whip up a 'brain to language-processing' mapping analogy that rivals the best of syntactic theories (pertaining to brain processing and phrase structure). Insofar as we will eventually be able to tease apart other non-linguistic processes that go on in the brain when language is generated, I am confident a whole new approach to syntactic theory is soon to be on the drawing board. I show optimism here because of the great strides that have been made in neuroimaging techniques. At the moment, brain scans such as ERPs, fMRI, PETs don't exclusively show language generation, but is rather mired in myriad non-linguistic factors—side-effects that, if not teased apart from the actual language stimuli, would also go into the equation confusing the target language stimuli and observed brain area activity. What we would like is a potential scan that represents the steps of pure language (without blurring outside noise such as motor-skill processing for phonology, decision making, criteria tasks such as word-to-semantic mappings, other engaged thinking, even day-dreaming). The request is for a filtered 'language-task' ERP. It's a tall order to fill. Recently new such scans have emerged, such as the MEG (Magnetoencephalography). The MEG works like the ERP scan in that neural events create the signal, an electrical current which flows down the dendrites of specific neurons that are active in the brain during the language task. This means that a strict one-to-one mapping may be available in representing the actual steps as language unfolds. Thus far, early studies show that verb processing indeed takes a dual-track approach:

- (i) *The first quarter second:* The Temporal-lobe (TL) parts of the brain are active with the search of lexical stems and irregular verbs. (As pointed out in the DMM, word stems and irregular verbs are stored and retrieved in associative memory). We extend this 'low-level' processing to incorporate the Verb Phrase (VP) as well since both *lexical words* and *VPs* would involve thematic/semantic relations. Hence, with this stage frozen as a 'snap-shot', we can theorize that children at the VP stage-one of syntactic development would only show TL activity with regards to MEG scans. (Wakefield and Wilcox (1995 p. 644), in support of Radford's (1990) maturational theory, believe that the TL and the LFT are initially unconnected in the first stages of language acquisition (Radford's VP stage-one). When set in correlation to the DMM, we view this stage absence of overt agreement/inflection and other functional material to be motivated by such brain computation deficits. As stated above, the specific deficits here would be due to a lack of early neurological myelination involved in linking the two areas of the brain. Since the FTL is not yet on-line in stage-one of language development, all generated language is seen as projecting from prosaic VP structures at TL).

By extension, we suggest that the acquisition of functional projections is triggered by the LFT being turned 'on' (analogous to a parameter setting) thus triggering the acquisition of functional morphology. In this sense, a two-prong conclusion can be

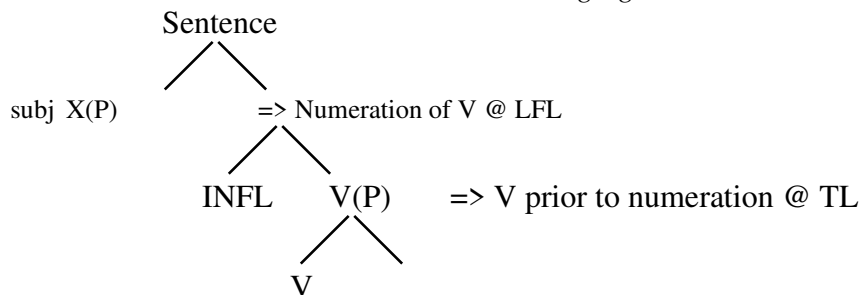
reached stating: while (i) we firmly believe that knowledge of morpho-phonology precedes and triggers syntax, (ii) such knowledge comes to the child via the computational maturation of that part of the brain that provides access to that specific knowledge in the first place.

By extension, we suggest children know that inflected forms are not lexical since inflected forms are conceptualized and generated in a separate mode of processing, disassociated from lexical stems. Such an analysis would go along way in supporting the independent work carried out by Elman and Newport's 'Less-is-more Hypothesis' by providing a mechanism by which an account for the 'starting small' can be made. Hence, we assume children start out without rules and paradigms.¹³ Only later are they said to *construct* their paradigms incrementally, one feature at a time, given the onset of LFT activity. It may be however that such inflectional languages which start out 'small' may otherwise use mechanisms available in other non-INFL languages (though still provided by UG) to mark notions of T/AGR. (See Hyams (1996) for a semantic treatments of otherwise null INFL markers.).

(ii) *A tenth of a second later:* The left/frontal-lobe (LFL) then becomes active with the pursuit of the regular verb functional affix (as pointed out by the DMM, affixes are rule-based). We extend this 'high-level' processing to incorporate the Inflectional Phrase (IP) as well since both affix formations and IPs involve abstract/rule-based relations. Hence, we would theorize that SLI subjects who show commission of functional errors (as discussed in §11 and in contrast to children in the 'no-functional' stage who show 'omission' of errors) would show LFL activity with regards to the MEG scans. Though word processing seems to be the order of the day, phrase processing is following right behind and gaining interest. As mentioned, if we can extend the lexical stem+affix split, as understood in the DMM, to that of the qualitative nature of different phrase types (IP vs. VP), then an even more robust theory of the DMM can be realized.

[86] Syntactic tree mapping inflectional neuroimaging

Syntactic Tree:

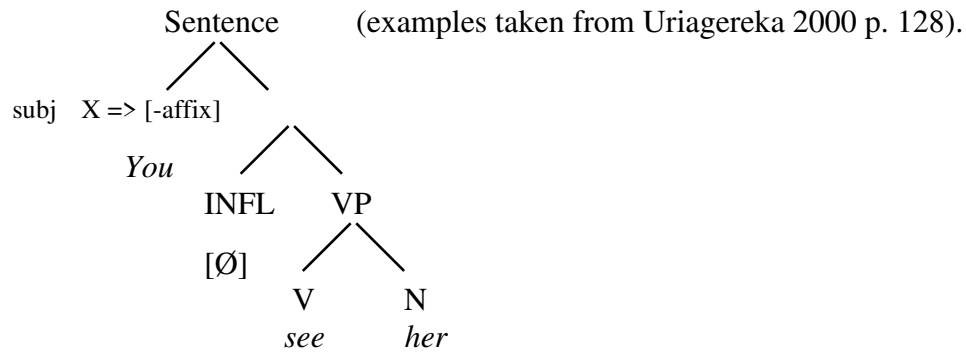
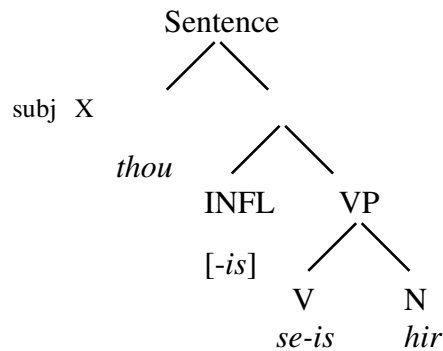


INFL = [Ø] when weak—covert movement of the verb stem into X.

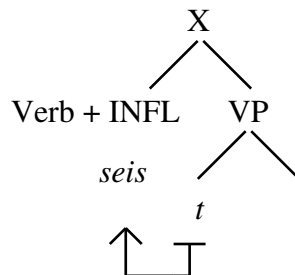
INFL = [affix] when strong—overt movement of the verb stem into X.

X = storage capacity #2—rule-based

V = storage capacity #1—associative

[87] *Weak/Covert movement: Modern English (single storage)*[88] *Strong/Overt movement: Middle English*

[89] Merge V to INFL/X:



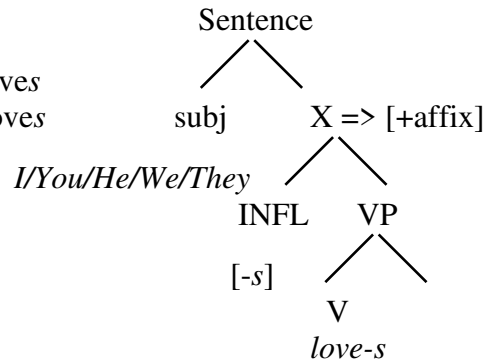
(Merge: Verb-stem {se} ‘pied-piping’ to INFL since the affix {-is} cannot stand alone at PF).

[90] *A Case of Strong Inflection: South-Western English INFL Paradigm*

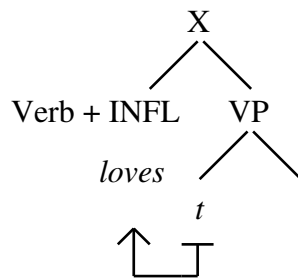
It seems that some varieties of English opt for a ‘rule-based’ storage capacity via a parameter setting. In stark contrast to Standard English, consider the rich inflectional paradigm for a variety of English spoken in South-Western England (Cheshire 1989, discussed in Radford 1997 p. 79).

Strong Tense/Agr features

I loves, You loves, He/She loves
We loves, You loves, They loves



[91] Merge V to INFL/X:



[92] There may be a maturational story behind all of this. How is it that some language types split-off the family tree and began taking on higher doses of abstract inflection? Perhaps '*ontogeny recapitulates phylogeny*' here in the sense that languages are (i) born from a Saussurean 'form-to-meaning' iconic system, (ii) then evolve into complex, less economical paradigmatic systems, (iii) only later to return to reduced paradigmatic systems. (The latter two stages are witnessed here to the extent that a shift has occurred from Middle English being strongly-inflected to Modern English being weakly-inflected). If so, then some languages may be simply lagging behind others on this evolutionary revolving door of 'simple-to-complex' inflectional typology. Of course, no one wants to claim that the brains of speakers of 'modular-complex' languages have somehow advanced over the others. The differences should rather be thought of as a sort of compensatory flow-sheet in that the brain is equipped to handle info in two fundamentally different ways—no one way is better than the other, there are different pay-offs (differing advantages to disadvantage). For example, no one wishes to claim that Chinese speakers have very little formal abstraction ability because their language demonstrates very little in the way of inflection. Rather, what should be said is that the language system itself has come to a nice equilibrium of sorts between the amount of abstraction that ought to be carried out (*efficiency*), which may be otherwise provided by the context in any account, and the amount of lexical storage required for communication without otherwise burdening the storage capacity (*effectiveness*). It is not clear, but Chinese may in fact require higher-levels of associative memory due to its internal structure. (All this is speculation at best; however, we can refer back to the footnote (6)

on Newport's hunch that associative storage capacity ultimately is responsible for the second capacity of abstraction). Noting such differences between, say, Spanish (with high amounts of inflection) and Chinese (with low to zero amounts of inflection), it needs to be understood that both languages have representations of functional categories (i.e., both have basically the same inflectional systems)—it is just that Spanish uses words and morphemes to represent these categories (=LFT storage), while Chinese does not. Instead, Chinese determines the presence of functional items such as Comp, Tense, Agreement via lexical context (=TL storage). Ultimately, it may be that the proposed parameterization leading to the typology amounts to little more than the following...

‘[T]here is evidence that [the] languages have basically the same inflectional systems, differing only in the way formal elements are accessed by the part of the computational procedure that provides instructions to articulatory perceptual organs. The mental computation seems otherwise identical, yielding indirect effects of inflectional structure that are observable, even if the inflections themselves are not heard in speech. That may well be the basis of language variation, in large measure’ (Chomsky 2000. p.120)

Notes

¹ See Clahsen et al. (2001). Ullman et al. (1997) suggest apparent processing distinctions reflect a *Language/Neural Dissociation* between Declarative (memory) and Procedural (rule) systems.

² Elements of this dual typology could be similarly interpreted along more traditional lines of typology—namely, ‘Semantic Transparent’ vs. ‘Grammaticization’-type languages (Hawkins 1986, Comrie 1987). Some attempts have been made to tease apart the two language types by using PET, and fMRI brain scans. Regarding some of these pioneering studies, there are early signs suggesting that the two language types may emphasize different parts of the brain for storage and retrieval tasks (typically equating to the Left-Frontal-Lobe (LFL) as the rule-based processing area vs. the Temporal-Lobe (TL) for frequency-based storage). Much more study needs to be done here however. But certainly, the idea that brain scan representations may come to model and map linguistic structure, akin to contemporary syntactic tree-diagramming, is an intriguing possibility. See Rodriguez-Fornells et al. 2002, Ullman et al. 1997, Posner & Raichle 1994, for brain imaging studies. See Neville et al. 1991, Hagoort et al. 1993 for research showing distinctive electro-physiological responses to syntactic vs. semantic violations. See Wakefield and Wilcox (1994) for a theoretical model of brain maturation and language acquisition. See Pinker (1999 pp. 264-268), Alegre & Gordon (1999), Clahsen et al. (2003) for general discussion.

³ Pinker (1999: 130) suggests that the proposed dual-track along with the single-track processing mode may be connected by an ‘inhibitory link’ by which the activation of a stored whole-word gradually slows down and ultimately turns off the rule. Strong-stem language types, which manifest very high whole-word frequency effects, might then be said to turn off the rule-based storage capacity. English irregular verbs and participle forms {-en} and {-ing} which turn off the rule-based storage could be thought of as maintaining ‘strong-stem-like’ properties. Posner et al. (p. 125-6) have discovered that PET language processing effects differ when the subject performs repetitive practice of a language task (see also fn 11). They claim practice turns off rule-based frontal-lobe processing and triggers areas in the Insular Cortex (temporal-lobe) region of the brain. Based on such findings, they suggest the brain has ‘two pathways’ for performing language generation tasks: (i) one used for automatic responses that were learned previously, and (ii) one

used for unlearned, but rule-based responses.

⁴ There is some thinking along these lines to suggest that the nature of the shift (or turning off) between the two processing storage capacities could be correlated to a critical mass of vocabulary storage. For instance, Marchman & Bates (1994) argue that children only begin to perform ‘weak-stem’ regular verb inflections once a critical mass of vocabulary has been reached showing a (weak-stem) regular to (strong-stem) irregular proportion of 55% to 45%. Likewise, Robinson & Mervis (1998) find similar correlates with onset of inflection to retardation of lexical growth. To my mind, there could be two opposing accounts of this: (i) that the eventual acquisition of a paradigm places a burden on the lexical memory apparatus thus triggering an additional storage capacity in the frontal lobe region of brain (a paradigm-to-memory relation), or conversely, (ii) that it is in fact a high memory capacity in the first place that forces the additional rule-based storage capacity (a memory-to-paradigm relation). (See Newport’s ‘Less-is-More Hypothesis’ for comments along this line of reasoning, along with subsequent work done by Jeff Elman on connectionism). In the latter view, high memory spawns rule-based paradigmatic function. Having said this, we must be clear here and state that we are in no way suggesting that ‘strong-stem’ language types suffer from a memory deficit which allows the language to maintain a single storage capacity. The idea is rather that inflectional modularity can be handled in two ways, depending on the nature of how the stem+affix enters into a processing and storage configuration.

⁵ Another way to analyze this within the Principles and Parameters Theory (Chomsky 1981, and later extended in the Minimalist Program, Chomsky 1995) would be to say that this ‘knowledge of morpho-phonology’ is akin to the kind of innate knowledge attributed to the invariant principles of UG, while the ‘triggering of syntax’ is more akin to variant parameter setting. In fact, both top-down and bottom up strategies eventually have to be involved: top-down for language typology which drives awareness in constructing an inflectional paradigm, bottom-up for constructing actual features in lexical items.

⁶ A good example of which could be ‘clitic affixation’ in the case of *can’t/shouldn’t/needn’t/etc.* where the negative {n’t} clitic attaches to the stem and then proceeds to be processed as part of the stem. The following examples only show licit movement of the neg clitic {n’t} particle across IP and into CP, and never the negation stem {not}—[*Shouldn’t*] *you finish your exam?* **Should you [n’t] finish your exams,* **Should not you finish your exams?* Such words that are derived at by clitic formation have been argued as being processed as morpho-phonological chunks. Also, derivational morphology which has some similarity to lexical semantics, likewise has been reported as maintaining ‘stem-like’ properties (see Anderson 1992: 184-5, Clahsen et al. 2001: 9).

⁷ Both languages appear to be described as weakly inflected [-INFL] like English which help define their ‘non-pro-drop’ status. Both French and German however, unlike English, are subject to overt syntactic finite verb movement at PF (V2 in German, and Verb IP-insertion in French).

⁸ For example, Bloom et al. (1980) note that early English production of [V+ing] was relegated to ‘activity’ verbs and was not initially generated on the full class of verbs. This suggests that early on at stage-1 children may process such inflections as a kind of lexical/semantic marker (data from Brown: 1973). In other studies, Santelmann & Jusczyk (1998) show that very young children can track dependency of the [V+ing] only up to limited syllabic separation (i.e., there is an Adjacency condition to such early [V+ing] constructs which point to semi-formulaic tendency). Such [V+ing] or [V+en] constructs may be based on high frequency or have a high repetitive nature. In addition, as uncovered by frequency effects obtained in lexical decision tasks,

high frequency may even affect otherwise rule-based constructions: Clahsen et al. (2003 p. 30) report that high frequency [verb+ed] constructs such as the verb *walked* may actually incorporate as a non-decomposed chunk and be stored/processed as a lexical subentry to the base form *walk* in the same way in which the irregular verb *drank* forms a subentry to *drink*. By contrast, the low frequency *stalked* showed no such subentry of *stalk*. Hence, high frequency of inflection may spawn memory versions.

⁹ See Clahsen et al. (2001). Ullman et al. (1997) suggest apparent processing distinctions reflect a *Language/Neural Dissociation* between Declarative (memory) and Procedural (rule) systems.

¹⁰ For discussion and data analysis of the early appearance of a ‘default’ DP containing a singular [+Interp] feature [+Def] in early child English grammar, see Galasso 2003c p.98). In brief, the discussion consists of the idea that lower syntactic positions dominated by VP may house +Interp features only and thus are not obliged to force movement of the head into a checking domain of a higher functional phrase (such as IP/AGRP). In this sense, Galasso (1999/2003c) treats the early emergence of DPs, which evidence no other signs of functional material, as a lower VP-dominated DP-projection with the sole semantic feature spell-out of [+Def] maintaining accusative/objective case by default. Examples of VP-dominated by VP would show DPs manifesting +Interp features only: e.g *Him kick* [VP/DP the ball] = [VP [/DP The ball] [V go]].

¹¹ Guasti and Rizzi (2002) suggest that indeed such bare stems are finite forms missing the {-s}, as indicated by the child sentence *He don’t hear me*’ (Sarah 3;5) which shows both NOM Case Agreement and potential Tense (only the Person feature is missing). However, if this is the case, it may ultimately throw the whole notion of phono-morphology into doubt, since in this view, it becomes impossible to tell whether or not a stem+affix actually correlates to any real morphological feature representation in the child’s grammar: If a [-FIN] stem shows Finiteness, what then prevents the potential and unwanted reading of a [+FIN] stem+affix showing no such finiteness? However, it also must be said that Meisel suggests just this when he posits that in Child German, it is the third person that activates the verbal paradigm and not the first and second person features.

¹² There is an interpretation that all functional categories and features are determined by UG and thus are universal: under this view, it is only their settings and specifications that must then be determined by the language input. The ‘strong maturational’ interpretation would view functional categories and features to be more tethered to maturational factors pertaining to the child’s brain development (akin to a biological basis of language development).

¹³ There is always a tempting aspect of the ‘Converging Theories Model’ to collapse and converge the ‘bottom-up’ cognitive Constructivist version of a stage-1 as based on high-frequency learning, slot-frame sequencing, lexical incorporation (cf. Tomasello 2000, Pine, Lieven and Rowland 1998) with that of the ‘top-down’ syntactic Generative version of the same stage-1 data (cf. Radford 1990, Radford and Galasso (1998), since both models descriptively overlap and call for an identical stage-1 void of any functional morphology. It may be possible that the DMM could converge the two positions in interesting ways as based on maturational brain development. For instance, and irrespective of whether or not the DMM is viable, children may simply be relying on the high frequency of lexical stems (in contrast to the consistent frequency of affixes) to determine the categorical distinction of lexical vs. function morphology. (A potential DMM ‘Convergence’ in this sense might mean that both Generativists’ and Constructivists’ hypotheses are basically correct for the initial stages of language acquisition, since both view the child’s language deficit in maturational terms).

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